

The 2-vs-3 Break-Even Frontier

Synthesis & Strategy — Workstream 4

A coaching briefing from the Pareto 2v3 shot-selection model

This is the project's final write-up, pulling together the break-even math from workstream 3, the population-level “Mega Graph” from workstream 2, and the zone-level frontiers from workstream 1 into one conclusion. It states the findings, not how they were built — working notes and methodology detail for all four workstreams live in the project's `report.md` and its decisions log.

Prepared for head coaches, analytics departments, and player-development teams.

The core relationship

A 2-point shot made at rate p_2 and a 3-point shot made at rate p_3 produce the same expected points when $2 \times p_2 = 3 \times p_3$ — so the break-even 2-point percentage, for any given 3-point percentage, is:

$$\text{FG2\%}_{\text{break-even}} = 1.5 \times \text{FG3\%}$$

Below that line, the average 2 is a worse bet than the average 3, points per shot. Above it, the reverse. This single relationship is the thread running through all three findings below.

FINDING 1

The league didn't clear its own break-even line until 2021-22

Applying the formula above to actual league-wide shooting each season turns up a specific, countable answer to “how the frontier has moved outward across eras” — and it's not the answer the modern “Moreyball” narrative would suggest.

League-wide 3-point percentage has been remarkably flat for three decades: 34–37% every single season from 1996-97 through 2025-26, no real trend up or down. What moved was the volume (3-point attempts rose from 21% of all field goal attempts in 1996-97 to 41–42% in the most recent two seasons — teams shooting roughly twice as many 3s per shot attempt as they did at the start of this dataset) and, more consequentially for this specific question, league-wide 2-point percentage: it climbed steadily, from 46–48% in the late 1990s to 54–55% in 2022–2026, as the long, contested mid-range jumper — the shot the “Moreyball” case was actually built against — was phased out of the league's shot diet in favor of shots at the rim.

The two lines (actual 2PT%, and $1.5 \times$ that season's 3PT%) don't cross until **2021-22**. For every season from 1996-97 through 2020-21, the league's actual 2-point percentage sat *below* its own break-even line — meaning that, in aggregate, the league would have scored more efficiently by taking more 3s and fewer 2s for essentially this entire span, even as teams were visibly moving in that direction the whole time. Only in the last handful of seasons has 2-point shooting improved enough, in aggregate, to actually justify the volume of 2s still being taken — and even then narrowly and not every season (2023-24 dipped back 0.2 points below the line).

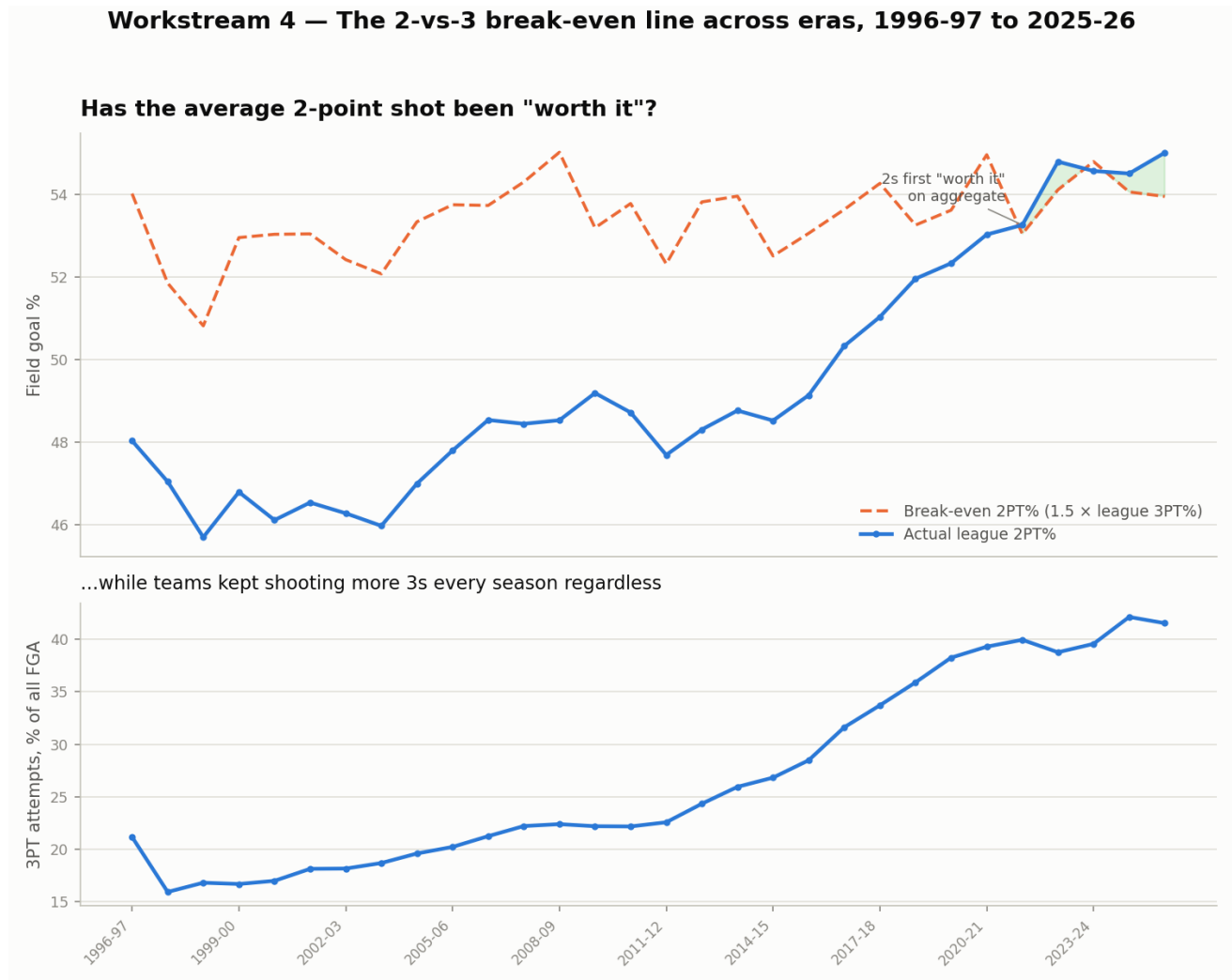


Figure 1. League-wide FG2% vs. that season's break-even line (top), and 3-point attempt rate (bottom), 1996-97 through 2025-26.

Source: data/breakeven_trend.csv, built by build_breakeven_trend.py from the same season totals workstream 2's Mega Graph uses.

The practical read: the league-wide shift toward 3-point volume over the last 15 years wasn't chasing a moving break-even target — the target barely moved, because 3-point percentage itself barely moved. It was teams improving the *quality* of the 2s they still took (fewer long twos, more rim shots) that eventually caught the break-even line up to where 3-point volume already was.

FINDING 2

“Pareto-efficient” is not the same as “a good shot”

Workstream 1's zone-level frontiers are the natural zone-level counterpart to Finding 1, and they surface a distinction worth stating plainly before drawing any coaching conclusion from them: a zone is “Pareto-efficient” here because *no other zone beats it on both expected points and risk at once* — not because it's a high-value shot in absolute terms. A low-risk, low-return zone can sit on the frontier perfectly legitimately, as the anchor at the safe end, without being a shot anyone should actually want more of.

That distinction matters because it's exactly what shows up in the non-RA (“jump shots only”) cut: **mid-range zones are Pareto-efficient 20 of 30 times** across the league average and the 5 named players — a surprisingly high rate, given the mid-range shot's reputation as the analytics era's designated inefficient shot. They're on the frontier because they're often the *lowest-risk* jump shot available, not because they're a good bet — every mid-range zone in this dataset has a lower expected value than the best available 3-point zone for that same player. The frontier includes them at the low-risk end the same way it includes Restricted Area at the low-risk end of the full frontier.

Two findings that *do* translate directly into “take more / fewer of these”:

- **“Above the Break 3” is never independently efficient once Restricted Area is in play** (0 of 18 full-frontier cells, across every entity) — it's always dominated by the rim. That's expected and not actionable on its own; it's the “duh, layups are good” restatement of the whole exercise.
- **Corner-3 is the strongest non-rim option, but only for shooters who can actually hit it.**
League-wide, both corners sit on the non-RA frontier with the highest expected value of any jump-shot zone. Per player, though, this splits hard: Stephen Curry's corner-3 numbers (1.57–1.67 points/shot) are elite and clearly worth hunting; Giannis Antetokounmpo has essentially no corner-3 shot at all — 3 attempts (all missed) from the left corner and none from the right in this dataset — not “inefficient” so much as “not part of his shot profile,” which is itself the finding for a player like him (see Finding 3).

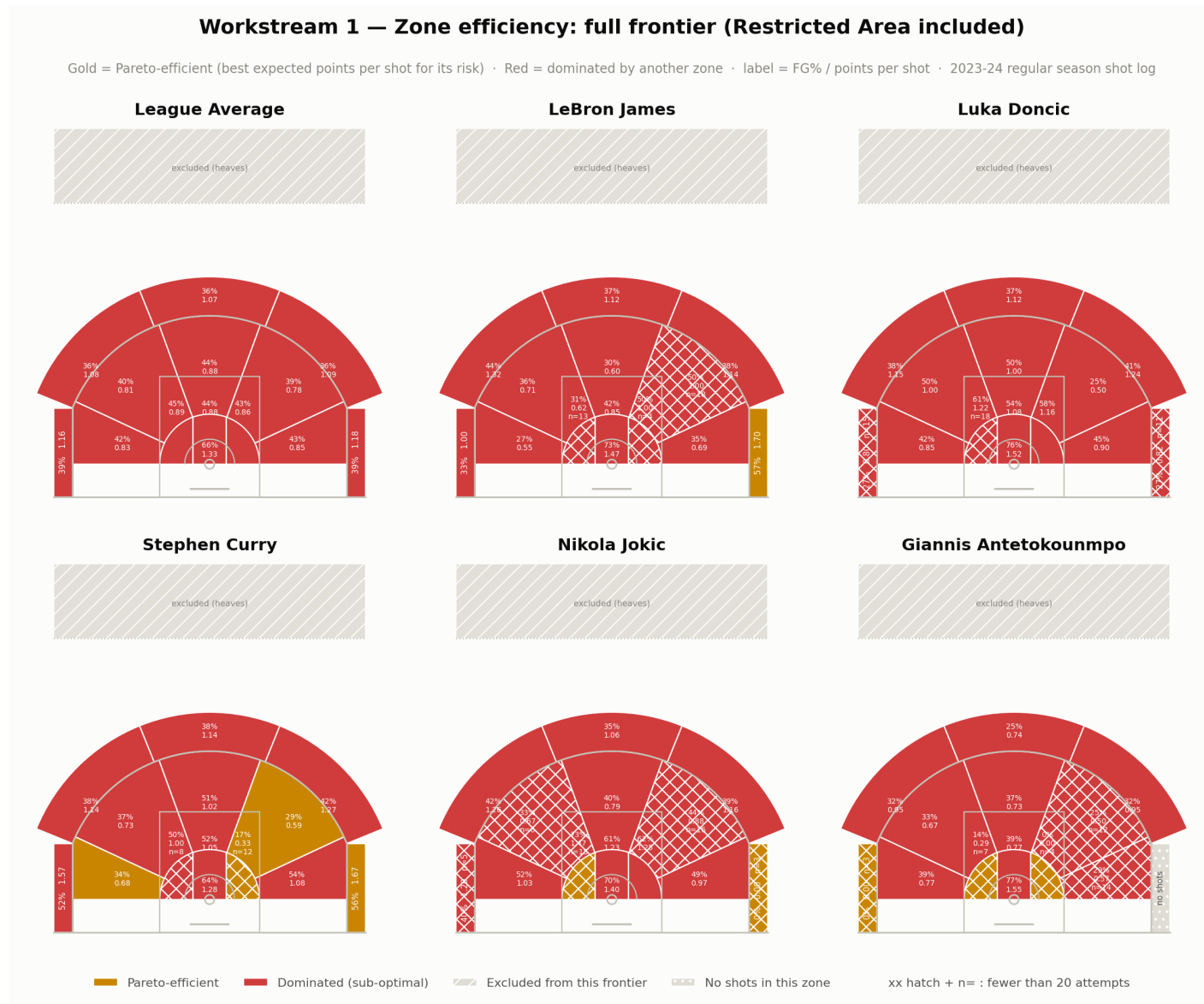


Figure 2. Zone-level Pareto frontier, full court — League Average and the 5 named players. Gold = efficient, red = dominated, gray = excluded (heaves).

Source: data/zone_efficiency.csv, 2023-24 shot-location data. Restricted Area included — the rim predictably dominates almost everything else.

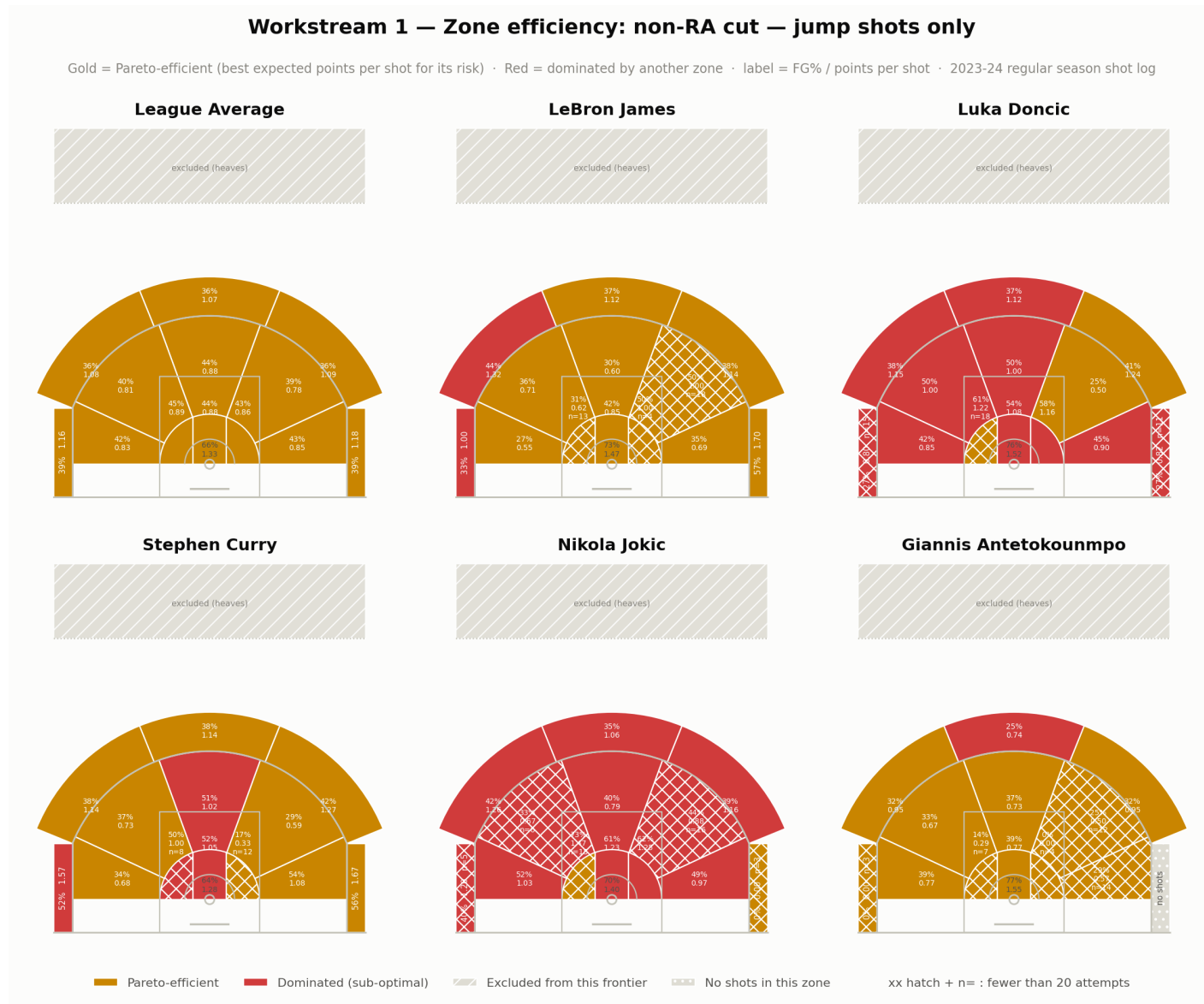


Figure 3. Zone-level Pareto frontier, non-RA cut — Restricted Area excluded to isolate the real jump-shot tradeoff among mid-range, corner-3, and above-the-break-3 zones.

Source: data/zone_efficiency.csv, 2023-24 shot-location data.

FINDING 3

The 5 players split into distinct shapes, not one archetype

- **Stephen Curry** is the one player in this group whose non-rim shot selection actually matches the league-wide “shoot more 3s” prescription at the individual level: his right-corner 3 (1.67 EV) is his single most valuable efficient zone, clearing even his own Restricted Area number (1.28) — a rarity in this dataset, where the rim usually tops every player's frontier outright. His mid-range zones (0.59–0.68) sit well below both. For him, the break-even math and his actual shot profile are already aligned.
- **LeBron James and Luka Doncic** both still get real value at the rim (1.47 and 1.52 EV) but differ from there. LeBron's right-corner 3 (1.70 EV, 23 attempts) actually *out-values* his own rim number and is Pareto-efficient on the full frontier right alongside it — a genuine “shoot more of this specific shot” finding. Luka's frontier is different: Restricted Area is his *only* full-frontier zone (every one of his 3-point zones is dominated once the rim is in the comparison), and only once Restricted Area is set aside does a best non-rim option emerge — right-side above-the-break 3 (1.24 EV). Two players who both finish well at the rim, but only one of them has a 3-point zone that stands on its own merits rather than just being “the best of what's left.”
- **Nikola Jokic** is the outlier finding of the whole workstream: his non-restricted-area paint shots (73% FG, 1.47 EV) *beat* his own Restricted Area number (70%, 1.40 EV) in this dataset — on only 15 attempts, flagged accordingly, but a real enough gap to be worth a team's own larger-sample follow-up rather than dismissing outright.
- **Giannis Antetokounmpo** is the clearest “one shot, everything else is a discount” profile of the five: Restricted Area (1.55 EV) towers over every zone he takes, and none of them are close — not because he barely shoots 3s (his above-the-break attempts, 120 combined across three sub-zones, are real volume) but because none of them work: 0.74–0.95 EV, all dominated. His corners are where “near-empty” actually applies — 3 attempts (all missed) from the left, none at all from the right in this dataset. There's no zone-efficiency argument for him to shoot more 3s from this data — the argument, if there is one, is about generating more shots at the rim in the first place, which is outside what a shot-selection frontier can speak to.

Caveats — read before acting on any of the above

- **League averages mask individual variance.** Finding 1 is a league-wide aggregate; it says nothing about whether *your* team's or *your* player's 2-point shooting has cleared its own break-even line. The same math applies at any level — team, player, even single game — but the numbers above are the league only.
- **Season mismatch between workstreams.** Finding 1 uses 1996-97 through 2025-26 season totals. Finding 2 and Finding 3 use 2023-24 shot-location data only — the one season with real zone-level data in this project. They're not describing the same year, and shouldn't be read as a single unified

season snapshot.

- **No defense, shot clock, or shot-creation difficulty in this model.** A “Pareto-efficient” zone is efficient conditional on a shot from there actually being available on a given possession — it says nothing about how hard that shot is to generate, who’s guarding it, or what happens to these numbers under playoff-level defensive attention.
- **Small samples are flagged, not fixed.** Several of the per-player zone numbers above (Jokic’s paint-left, LeBron’s several under-20-attempt zones, both of Giannis’s corner-3 splits) rest on fewer than 20 shots for that player-zone combination in a single season. They’re real numbers from real data, not typos or bugs, but they carry real sampling uncertainty that a full season-over-season average would substantially narrow.
- **The portfolio framing is a simplification.** Treating each shot as an independent, one-shot risk/return draw is the framing the whole project is built on, and it’s a reasonable one for this kind of analysis — but it doesn’t capture second-order effects like offensive rebounding rates by shot type, free-throw generation, or how a team’s shot profile affects its opponent’s transition opportunities.

Where this leaves the four workstreams

1	Efficient Zones + Players	Done — zone-level Pareto heatmaps
2	Mega Graph + Efficiency Gap	Done — static + interactive scatter plots
3	2 vs 3 (core scatter + break-even relation)	Done — folded into workstreams 1, 2, and 4
4	Synthesis & Strategy	Done — this document

All four are now built. `report.md` remains the working methodology notes; the project decisions log has the full session-by-session history of what was tried, what changed, and why.